

Data Mining

(DS3005/DS3002)

BDS-6A/6B/6C/8A

Date: Apr 09th 2026

Course Instructor: Dr. Tahir Ejaz

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 25

Total Questions: 3

Student Signature

1. Attempt all the questions.
2. Rough sheets are *allowed* but you are required to write *solution* in the provided space.
Do not submit the rough sheets.

CLO # 1: Understand basic concepts of data mining.

Q1:

Following table shows the True Positive Rate (column 2) against the probability value returned by a probabilistic classifier for each of the 7 tuples in a test set (the tuples are sorted in the decreasing probability order). Fill in the value of False Positive Rate (column 3) against each tuple. [3.5 Marks]

Tuple #	TPR	FPR
1	0.2	0.8
2	0.4	0.6
3	0.6	0.4
4	0.6	0.4
5	0.8	0.2
6	0.8	0.2
7	1.0	0

~~FF~~
FPR = 1 - TPR

Illustrate a situation with 5 tuples (along with their actual class labels and the probability values returned by a probabilistic classifier) where AUC of the model is 1. [1 Mark]

Tuple #	Class	Probability
1 7	True	100%
2 6	True	80%
3 3	True	60%
4 4	True	60%
5 5	True	80%

In general what condition/s must be met, so that AUC of a model is 1.

[0.5 Marks]

The TPR should be away from diagonal value

and area under should be 1 to be a model with AUC 1-

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CLO # 1: Understand basic concepts of data mining.

Q2: Using the data given below, calculate the Information Gain and Gain Ratio of each feature. [1+3*3 Marks]

Outlook	Temp	Windy	Play Golf
Rainy	Hot	False	No
Rainy	Hot	True	No
Overcast	Hot	False	Yes
Sunny	Mild	False	Yes
Sunny	Cool	False	Yes
Sunny	Cool	True	No
Overcast	Cool	True	Yes
Rainy	Mild	False	No
Rainy	Cool	False	Yes
Sunny	Mild	False	Yes

Show proper working!

$$Info(D) = -\frac{4}{10} \log_2 \frac{4}{10} - \frac{6}{10} \log_2 \frac{6}{10} = 0.971$$

$$I(3,1) = -\frac{3}{4} \log_2 \left(\frac{3}{4}\right) - \frac{1}{4} \log_2 \left(\frac{1}{4}\right) = 0.811$$

Info_{Outlook}(D):

$$= \frac{4}{10} \times I(3,1) + \frac{2}{10} \times I(2,0) + \frac{4}{10} \times I(3,1)$$

$$= \frac{4}{10} \times 0.811 + \frac{2}{10} \times 0 + \frac{4}{10} \times 0.811 = 0.6488$$

Gain (Outlook):

$$= 0.971 - 0.6488$$

$$= 0.3222$$

SplitInfo_{Outlook}(D):

$$= -\frac{4}{10} \log_2 \left(\frac{4}{10}\right) - \frac{2}{10} \log_2 \left(\frac{2}{10}\right) - \frac{4}{10} \log_2 \left(\frac{4}{10}\right)$$

$$= 1.5219$$

GainRatio (Outlook):

$$= \frac{0.3222}{1.5219} = 0.2117$$

Info_{Temp}(D): $I(2,1) = -\frac{2}{3} \log_2 \left(\frac{2}{3}\right) - \frac{1}{3} \log_2 \left(\frac{1}{3}\right) = 0.9183$

$$Temp = \frac{3}{10} \times 0.9183 + \frac{3}{10} \times 0.9183 + \frac{4}{10} \times 0.811 = 0.87578$$

Gain (Temp): $= 0.971 - 0.87578$

$$= 0.09562$$

SplitInfo_{Temp}(D):

$$= -\frac{3}{10} \log_2 \left(\frac{3}{10}\right) - \frac{3}{10} \log_2 \left(\frac{3}{10}\right) - \frac{4}{10} \log_2 \left(\frac{4}{10}\right)$$

$$= 1.57$$

GainRatio (Temp):

$$= \frac{0.09562}{1.57} = 0.0609$$

Info_{Windy}(D): $I(2,5) = -\frac{5}{7} \log_2 \left(\frac{5}{7}\right) - \frac{2}{7} \log_2 \left(\frac{2}{7}\right) = 0.86$

$$= \frac{3}{10} \times 0.9183 + \frac{7}{10} \times 0.86 = 0.87749$$

Gain (Windy):

$$0.971 - 0.87749 = 0.09351$$

SplitInfo_{Windy}(D):

$$= -\frac{3}{10} \log_2 \left(\frac{3}{10}\right) - \frac{7}{10} \log_2 \left(\frac{7}{10}\right) = 0.8813$$

GainRatio (Windy):

$$= \frac{0.09351}{0.8813} = 0.106$$

CLO # 2: Apply data mining techniques to extract information from large data sets.

Q3: Using the Naïve Bayesian classification algorithm, calculate the marginal probability (i.e. $P(x)$), where x is:

Outlook	Temp	Windy
Rainy	Mild	True

As intermediate steps, also calculate the likelihoods (i.e. $P(x | \text{Yes})$ and $P(x | \text{No})$). Show proper working!

Note: Use the data given in Question 2.

[4+4+2 Marks]

$P(x | \text{Yes})$:

	Yes	No
Rainy	1	3
Mild	2	1
True	1	2

$$= \frac{1}{6} \times \frac{2}{6} \times \frac{1}{6} \times \frac{6}{10}$$

$$= 0.0055$$

$$\Rightarrow \frac{0.0055}{0.043} = \boxed{0.128}$$

$$\boxed{12.8\%}$$

$P(x | \text{No})$:

$$= \frac{3}{4} \times \frac{1}{4} \times \frac{2}{4} \times \frac{4}{10} = 0.0375$$

$$\Rightarrow \frac{0.0375}{0.043} = \boxed{0.872}$$

$$\boxed{87.2\%}$$

$P(x)$:

Hint: Recall the total probability theorem.

$$0.0055 + 0.0375$$

$$= \boxed{0.043}$$